US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12405014
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Kim; Junhee et al.

Outdoor unit of air conditioner

Abstract

An outdoor unit of an air conditioner may include a base panel, a front panel disposed on a front end of the base panel and having a through hole formed therein, an orifice fitted to the through hole, a fan disposed behind the through hole and having at least a portion accommodated inside of the orifice, and a heat exchanger disposed on or at an edge of a rear side of the base panel. A frontward-to-backward width of a first side end of the orifice may be less than a frontward-to-backward of a second side end of the orifice.

Inventors: **Kim; Junhee (Seoul, KR), Oh; Siyoung (Seoul, KR), Kim; Hyoungjune (Seoul, KR), Choi; Siyoung (Seoul, KR), Lee; Jonghae (Seoul, KR), Lee; Gwansoo (Seoul, KR), Park; Kiyoon (Seoul, KR), Bae; Hyunwook (Seoul, KR), Kim; Taegyu (Seoul, KR)**

Applicant: **LG ELECTRONICS INC.** (Seoul, KR)

Family ID: **88297082**

Assignee: **LG ELECTRONICS INC. (Seoul, KR)**

Appl. No.: **18/502311**

Filed: **November 06, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20250012459 A1	Jan. 09, 2025

Foreign Application Priority Data

KR	10-2023-0086994	Jul. 05, 2023

Publication Classification

Int. Cl.: **F24F1/38** (20110101); **F04D29/52** (20060101); **F04D29/54** (20060101); **F24F1/40** (20110101)

U.S. Cl.:

CPC **F24F1/38** (20130101); **F04D29/526** (20130101); **F04D29/541** (20130101); **F24F1/40** (20130101);

Field of Classification Search

CPC: F24F (1/38); F04D (29/526); F04D (29/541)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6450760	12/2001	Furukawa	416/189	F04D 29/384
9062888	12/2014	Tadokoro	N/A	F04D 29/545
9683580	12/2016	Tyler	N/A	F01P 11/10
9829010	12/2016	Yoshida	N/A	F04D 29/164
11009042	12/2020	Takeuchi	N/A	F04D 29/164
11181125	12/2020	Chang	N/A	F04D 25/0646
11209011	12/2020	Kim	N/A	F24F 1/38
2007/0280829	12/2006	Stevens	416/189	F04D 29/164
2010/0199716	12/2009	Murakami et al.	N/A	N/A
2010/0269537	12/2009	Tadokoro et al.	N/A	N/A
2013/0142460	12/2012	Ng	N/A	N/A
2016/0341220	12/2015	Kondou	N/A	F04D 29/164
2017/0167773	12/2016	Jeon	N/A	N/A
2017/0350412	12/2016	Hioki	N/A	F04D 29/5833

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2175214	12/2009	EP	N/A
2009-203980	12/2008	JP	N/A
4690682	12/2010	JP	N/A
10-2007-0046463	12/2006	KR	N/A
10-1734722	12/2016	KR	N/A
10-2020-0043847	12/2019	KR	N/A
10-2022-0078299	12/2021	KR	N/A
WO 2009/0113338	12/2008	WO	N/A

OTHER PUBLICATIONS

Machine translation of KR20200043847 (Year: 2025). cited by examiner
Machine translation of JP2009203980 (Year: 2025). cited by examiner
Optimization of flow performance by designing orifice shape of outdoor unit of air-conditioner, Seo-Yoon Ryu et al., The Journal of the Acoustical Society of Korea vol. 36, No. 6 (2017). cited by applicant

Korean Office Action dated Dec. 11, 2024 issued in Application No. 10-2023-0086994. cited by applicant
European Search Report issued in Application No. 23202391.1 dated Mar. 11, 2024. cited by applicant

Primary Examiner: Zerphey; Christopher R

Attorney, Agent or Firm: KED & ASSOCIATES

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)
(1) The present application claims the benefits of priority to Korean Patent Application No. 10-2023-0086994, filed in Korea on Jul. 5, 2023, the contents of which is incorporated herein by reference in its entirety.
BACKGROUND
1. Field
(2) An outdoor unit of an air conditioner is disclosed herein.
2. Background
(3) An air conditioner is a device that cools and heats an indoor space through heat exchange between a refrigerant flowing through a heat exchange cycle and indoor air and outdoor air. More specifically, the air conditioner includes a compressor that compresses the refrigerant, an outdoor heat exchanger that exchanges heat between the refrigerant and outdoor air, and an indoor heat exchanger that exchanges heat between the refrigerant and indoor air.
(4) The air conditioner may be equipped with a heat storage tank. In the heat storage tank, a fluid heated or cooled by the refrigerant circulating in the air conditioner may be stored. For example, the fluid may include water.
(5) The outdoor unit of the air conditioner includes a case, the compressor accommodated inside of the case, the outdoor heat exchanger, and the heat storage tank. An orifice is provided on a front surface of the outdoor unit, and a fan is provided between the orifice and the outdoor heat exchanger. The fan may be an axial fan. The orifice forms a flow path for air forced to flow by the axial fan, and guides the air, which has passed through the outdoor heat exchanger, to flow in an axial direction of the fan and to be discharged outside of the outdoor unit.
(6) Air passing through the orifice generates noise as it passes through blades of the fan. The noise is generated by formation of a vortex in a gap formed between an end of the blade and an inner circumferential surface of the orifice. As flow resistance is generated by the vortex, there is a disadvantage in that the air that has passed through the outdoor heat exchanger cannot be quickly discharged outside of the outdoor unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Embodiments will be described in detail with reference to the following drawings in which like reference numerals refer to like elements, and wherein:
(2) FIG. **1** is a perspective view of an outdoor unit configuring an air conditioner equipped with a heat storage tank according to an embodiment;
(3) FIG. **2** is an exploded perspective view showing a state of coupling an orifice and a front panel

according to an embodiment;
(4) FIG. **3** is a rear view of the orifice of FIG. **2**;
(5) FIGS. **4** and **5** are enlarged views of a portion A of FIG. **1**, showing a state in which a fastening hook of the orifice is inserted into a hook groove of the front panel according to an embodiment;
(6) FIG. **6** is a cross-sectional view of the orifice, taken along line VI-VI of FIG. **2**; and
(7) FIG. **7** is an enlarged cross-sectional view of a portion B of FIG. **1**.

DETAILED DESCRIPTION

(8) Hereinafter, an outdoor unit of an air conditioner according to an embodiment will be described with reference to the drawings. Wherever possible, like reference numerals have been used to indicate like components, and repetitive disclosure has been omitted.
(9) FIG. **1** is a perspective view of an outdoor unit configuring an air conditioner equipped with a heat storage tank according to an embodiment. Referring to FIG. **1**, outdoor unit **10** of the air conditioner according to an embodiment may include a base panel **11**, a front panel **12** disposed at a front end of the base panel **11**, an orifice **13** coupled to the front panel **12**, a fan **15** disposed behind the orifice **13**, and a heat exchanger **14** disposed on an upper surface of the base panel **11**.
(10) The outdoor unit **10** may further include a barrier **16** disposed on an upper surface of the base panel **11**. The barrier **16** may be understood as a partition that partitions an upper space of the base panel **11** into a heat exchange space on a left or first side and an electrical equipment space on a right or second side. In addition, the heat exchanger **14** may be erected in the heat exchange space and be bent and extend along side and rear ends of the base panel **11** to define side and rear surfaces of the heat exchange space.
(11) One or a first side end of the front panel **12** may be coupled to a front end of the heat exchanger **14**, and the other or a second side end may be coupled to a front end of the barrier **16**. A rear end of the barrier **16** may be connected to a side end of the heat exchanger **14**.
(12) The outdoor unit **10** may further include a compressor **19** disposed on the upper surface of the base panel **11** corresponding to the electrical equipment space, a heat storage tank **20**, and a control box **17** located above the heat storage tank **20**. The heat storage tank **20** may be spaced apart from the compressor **19**, and a flow path may be provided inside of the heat storage tank **20** to enable heat exchange between refrigerant and water without mixing.
(13) When the fan **15** rotates, outside air flows into the outdoor unit **10** through a short side of the heat exchanger **14** forming a side surface of the outdoor unit **10** and a long side of the heat exchanger **14** forming a rear surface of the outdoor unit **10**. Air, which has passed through the heat exchanger **14**, flows from a rear surface of the fan **15** toward a front surface thereof, passes through the orifice **13**, and is then discharged outside of the outdoor unit **10**. At least some of a plurality of blades **151** of the fan **15** are accommodated inside of the orifice **13** so that all air forced to flow by the fan **15** passes through the orifice **13**.
(14) FIG. **2** is an exploded perspective view showing a state of coupling an orifice and a front panel according to an embodiment. FIG. **3** is a rear view of the orifice of FIG. **2**.
(15) Referring to FIGS. **2** and **3**, the front panel **12** to which the orifice **13** is coupled may include a panel body **121** in the form of a rectangular plate. The panel body **121** may have a through hole **1201** formed therein. The orifice **13** may be fitted into and coupled to the through hole **1201**.
(16) More specifically, a fastening sleeve **122** that fastens the orifice **13** to the front panel **12** may extend from an edge of the through hole **1201**. The fastening sleeve **122** may be bent toward a front side of the front panel **12** and contact a side surface of the orifice **13**.
(17) A plurality of fastening steps **1222** may be formed in the fastening sleeve **122**, and a screw hole **1223** may be formed in each fastening step **1222**. For example, the plurality of fastening steps **1222** may be formed on upper left and right (first and second) sides and lower left and right (third and fourth) sides, respectively, based on a vertical line bisecting the through hole **1201** into left and right (first and second) sides.
(18) A misassembly prevention groove **1224** may be formed in the fastening sleeve **122**

corresponding to a space between adjacent fastening steps. The misassembly prevention groove **1224** may be formed in upper and lower ends of the fastening sleeve **122**. In addition, a pair of misassembly prevention grooves **1224** may be formed on the left or right side of the vertical line bisecting the through hole **1201**.

(19) One or more hook grooves **1221** may be formed in the fastening sleeve **122**. For example, the hook grooves **1221** may be formed at positions facing each other on the left and right sides of the through hole **1201**.

(20) A coupling guide **1226** may be formed on the front surface of the front panel **12**. The coupling guide **1226** functions to guide the orifice **13** to be correctly coupled to or at a correct position, and may be formed at any point between an upper end of the through hole **1201** and an upper end of the panel body **121**. The coupling guide **1226** may have an inverted triangle shape. Also, a portion of the panel body **121** may protrude forward due to a forming process.

(21) The orifice **13** may include a panel seating portion **131** configured to be in close contact with the front surface of the front panel **12**, a diffuser **132** rounded backward from the panel seating portion **131**, an inducer **134** that extends roundly backward from a rear end of the diffuser **132**, a vortex fence **133** that surrounds the inner circumferential surface of the orifice **13** corresponding to a boundary between the diffuser **132** and the inducer **134**, and through-hole **1301**.

(22) More specifically, the rear end of the inducer **134** may be defined as the rear end of the orifice **13**, and the panel seating portion **131** may be defined as a front surface of the orifice **13**. In addition, the diffuser **132** may include a rounded portion rounded such that a diameter thereof decreases as it extends rearward from a rear end of the panel seating portion **131**, and a straight portion that extends rearward from a rear end of the rounded portion. In addition, the inducer **134** may include a straight portion that extends rearward from the straight portion of the diffuser **132** and a rounded portion that is rounded in a direction in which a diameter thereof increases as it extends rearward from a rear end of the straight portion.

(23) With this structure, the orifice **13** may be described a cylindrical body including a front extension (diffuser **132**) that is rounded forward from a front end of the body, a rear extension (inducer **134**) that is rounded rearward from a rear end of the body, and a close contact portion (panel seating portion **131**) that extends from a front end of the front extension and comes into close contact with the front surface of the front panel **12**. In addition, the vortex fence **133** may protrude from an inner circumferential surface of the body and surround it in a band shape.

(24) A plurality of reinforcing ribs **136** may protrude from an outer circumferential surface of the orifice **13**. The plurality of reinforcing ribs **136** may be spaced apart in a circumferential direction of the orifice **13**.

(25) Fastening hooks **135** may protrude from a back surface of the panel seating portion **131**. The fastening hooks **135** may be inserted into the hook grooves **1221** of the front panel **12**. Accordingly, the fastening hooks **135** may also be formed on the left and right sides of the orifice **13**, respectively.

(26) A plurality of fastening steps **137** may be formed in or at the front surface of the orifice **13**, and a screw hole **1371** may be formed in each fastening step **137**. The plurality of fastening steps **137** may be formed at positions corresponding to the plurality of fastening steps **1222** formed on the front panel **12**, respectively. That is, the plurality of fastening steps **137** may be formed on the left and right sides of the front upper and lower ends of the orifice **13**, respectively.

(27) A coupling guide **138** may be formed at an upper end of the panel seating portion **131**. The coupling guide **138** may be formed at a position facing the coupling guide **1226** formed on the front panel **12**, and may have an equilateral triangle shape; however, embodiments are not limited thereto.

(28) More specifically, when the orifice **13** is completely coupled to the front panel **12**, the coupling guides **138** and **1226** may be located on a same vertical line. The coupling guides **138** and **1226** prevent upper and lower surfaces of the orifice **13** from being reversed and combined. More

specifically, as left and right portions of the orifice **13** according to embodiments form an asymmetrical shape, there is a limitation in that the upper and lower portions should not be reversed and combined. To prevent such misassembly, the coupling guides **138** and **1226** are provided.
(29) As an additional unit for preventing misassembly, misassembly prevention ribs may be formed on the back surface of the orifice **13**. The misassembly prevention ribs may include upper misassembly prevention rib **1391** formed on an upper side of the back surface of the orifice **13** and lower misassembly prevention rib **1392** formed on a lower side of the back surface. The upper and lower misassembly prevention ribs **1391** and **1392** may be respectively fitted into the upper and lower misassembly prevention grooves **1224** and **1225** of the front panel **12**. Accordingly, the upper and lower misassembly prevention ribs **1391** and **1392** may also be formed at points spaced to the left or right from the vertical line bisecting the orifice **13**.
(30) FIGS. **4** and **5** are enlarged views of a portion A of FIG. **1**, showing a state in which the fastening hook of the orifice is inserted into the hook groove of the front panel. Referring to FIG. **4**, the fastening hooks **135** are formed on the left and right sides of the orifice **13**, respectively, and are formed symmetrically with respect to a horizontal line. That is, the fastening hook **135** on the right may extend downward clockwise in the drawing, and the fastening hook **135** on the left may extend upward clockwise.
(31) More specifically, when the orifice **13** is in close contact with the front surface of the front panel **12**, a lower end of the fastening hook **135** is fitted into the hook groove **1221**. In this state, when the orifice **13** rotates clockwise, the fastening hook **135** on the right side rotates from an upper end to a lower end of the hook groove **1221** and is caught in the lower end of the hook groove **1221**. Conversely, the fastening hook **135** on the left rotates from the lower end to the upper end of the hook groove **1221** and is caught in the upper end of the hook groove **1221**. The structure of the fastening hook **135** and the hook groove **1221** may reduce the number of fastening steps **1222** and **137** for screw coupling, thereby shortening an assembly time and reducing manufacturing costs.
(32) FIG. **6** is a cross-sectional view of the orifice, taken along line VI-VI of FIG. **2**. Referring to FIG. **6**, left and right (first and second) ends of the orifice **13** according to embodiments are designed to have different widths in a frontward-to-backward direction.
(33) More specifically, the left portion of the orifice **13** adjacent to a short side of the heat exchanger **14** and the right portion of the orifice **13** adjacent to the barrier **16** are designed to have different shapes. More specifically, the larger the frontward-to-backward width of the orifice **13**, the larger an area for accommodating the blades **151** of the fan **15**, which has the advantage of increasing air volume. However, if the width of the left portion of the orifice **13** adjacent to the short side of the heat exchanger **14** increases, outside air suctioned through the short side of the heat exchanger **14** may flow into the orifice **13**, and there is a disadvantage in that flow resistance increases. In order to overcome this problem, the frontward-to-backward width L**2** of the right portion of the orifice **13** is designed to be larger than the frontward-to-backward width L**1** of the left portion of the orifice **13**.
(34) Accordingly, a left rear end of the orifice **13** is formed to be tapered so that the frontward-to-backward width increases linearly from the left end of the orifice **13** to a center of the orifice **13**, as shown. A right rear end of **13** of the orifice **13** is formed parallel to the front surface of the orifice **13**.
(35) FIG. **7** is an enlarged cross-sectional view of a portion B of FIG. **1**. Referring to FIG. **7**, the fan **15** is provided in a form in which at least some of the blades are accommodated inside of the orifice **13**, so that all air flowing by the fan **15** passes through the orifice **13**.
(36) More specifically, the blade **151** of the fan **15** is located at a rear side of the vortex fence **133**, and outer diameter R**2** of the blade **151** may be designed to be larger than or equal to inner diameter R**1** of the vortex fence **133**.

(37) As the suctioned air passes through a gap formed between the end of the blade **151** and the inner circumferential surface of the orifice **13**, a vortex is formed. As the vortex fence **133** protrudes, formation of the vortex is suppressed or prevented, air volume increases, and noise is reduced.
(38) The vortex fence **133** may include a straight portion **1331** that extends radially from the inner circumferential surface of the orifice **13**, and a rounded portion **1332** rounded from an upper end of the straight portion **1331** toward the inner circumferential surface of the orifice **13**. The suctioned air passes over the rounded portion **1332** and is discharged to a front side of the outdoor unit **10**.
(39) Embodiments disclosed herein solve at least the above-discussed problems.
(40) An outdoor unit of an air conditioner according to embodiments disclosed herein may include a base panel, a front panel erected or disposed on a front end of the base panel and having a through hole formed therein, an orifice fitted to the through hole, a fan placed or disposed behind the through hole and having at least a portion accommodated inside of the orifice, and a heat exchanger placed or disposed on or at an edge of a rear side of the base panel. A frontward-to-backward width of one or a first side end of the orifice is less than that of the other or a second side end of the orifice. The heat exchanger may include a short side that extends along one or a first side end of the base panel and a long side bent from a rear end of the short side and extending along a rear end of the base panel, one end of the orifice may be adjacent to the short side, and the frontward-to-rearward width of the orifice may increase linearly from one end of the orifice to a center of the orifice.
(41) The orifice may include a cylindrical body, a front extension rounded forward from a front end of the body, a close contact portion that extends from a front end of the front extension and configured to come into close contact with a front surface of the front panel, and a rear extension rounded rearward from a rear end of the body. A vortex fence may protrude from an inner circumferential surface of the body to be formed in a band shape. A rear surface of the vortex fence may be formed to be rounded with a predetermined curvature.
(42) An end of a blade forming the fan may be located behind the vortex fence. An outer diameter of the blade may be equal to or greater than an inner diameter of the vortex fence.
(43) Coupling guides may be formed on a front surface of the front panel and the close contact portion, respectively. The coupling guides may be placed or disposed on a same vertical line in a state in which the orifice is completely coupled to the front panel
(44) A fastening sleeve may extend at or from an edge of the through hole. At least a portion of the fastening sleeve may be bent to or toward a front side of the front panel.
(45) The outdoor unit according to embodiments disclosed herein may further include misassembly prevention ribs formed at upper and lower ends of a back surface of the front panel, respectively, and misassembly prevention grooves formed in or at upper and lower ends of the fastening sleeve, respectively, to accommodate the misassembly prevention ribs. The misassembly prevention ribs and the misassembly prevention grooves may be formed at points spaced to the left or right from a vertical line passing through a center of the through hole.
(46) The outdoor unit according to embodiments disclosed herein may further include fastening hooks that protrude from a back surface of the close contact portion, and hook grooves formed in the fastening sleeve to accommodate the fastening hooks. When the orifice rotates in a state of being in close contact with the front panel, the fastening hooks are caught in the hook grooves.
(47) The fastening hooks may protrude from left and right sides of a back surface of the close contact portion, respectively. The hook grooves may be formed in left and right sides of the fastening sleeve, respectively, and the pair of fastening hooks may be formed symmetrically with respect to a horizontal plane.
(48) The outdoor unit of the air conditioner according to embodiments disclosed herein configured as described above has at least the following advantages.
(49) First, a cross-section of the orifice has an asymmetric structure, which has the effect of

increasing air volume.
(50) Second, a vortex fence that suppresses or prevents vortex generation protrudes from the inner circumferential surface of the orifice, which has the effect of increasing air volume and reducing noise.
(51) Third, by improving an assembly structure between the orifice and a front panel, there is an advantage in reducing the number of screws to secure the orifice to the front panel.
(52) Fourth, there is an advantage in preventing misassembly of the orifice by providing a misassembly prevention unit to prevent the orifice having an asymmetric shape from being incorrectly assembled on the front panel.
(53) It will be understood that when an element or layer is referred to as being “on” another element or layer, the element or layer can be directly on another element or layer or intervening elements or layers. In contrast, when an element is referred to as being “directly on” another element or layer, there are no intervening elements or layers present. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.
(54) It will be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section could be termed a second element, component, region, layer or section without departing from the teachings of the present invention.
(55) Spatially relative terms, such as “lower”, “upper” and the like, may be used herein for ease of description to describe the relationship of one element or feature to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “lower” relative to other elements or features would then be oriented “upper” relative to the other elements or features. Thus, the exemplary term “lower” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.
(56) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.
(57) Embodiments are described herein with reference to cross-section illustrations that are schematic illustrations of idealized embodiments (and intermediate structures). As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.
(58) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.
(59) Any reference in this specification to “one embodiment,” “an embodiment,” “example

embodiment," etc., means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of such phrases in various places in the specification are not necessarily all referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with any embodiment, it is submitted that it is within the purview of one skilled in the art to effect such feature, structure, or characteristic in connection with other ones of the embodiments.

(60) Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. An outdoor unit of an air conditioner, comprising: a base panel; a front panel disposed at a front end of the base panel and having a through hole formed therein; an orifice fitted to the through hole; a fan disposed behind the through hole, wherein at least a portion of the fan is accommodated inside of the orifice; and a heat exchanger disposed on an edge of a rear side of the base panel, wherein a frontward-to-backward width of a first side end of the orifice is less than a frontward-to-backward of a second side end of the orifice, wherein the heat exchanger comprises: a first side that extends along a first side end of the base panel; and a second side bent from a rear end of the second side and that extends along a rear end of the base panel, wherein the first side is shorter than the second side, wherein the first side end of the orifice is adjacent to the first side, wherein the frontward-to-backward width of the orifice increases linearly from the first side end of the orifice to a center of the orifice and is formed to extend parallel to a front surface of the orifice from the center of the orifice to the second side end of the orifice, wherein a fastening sleeve extends from an edge of the through hole of the front panel, wherein at least a portion of the fastening sleeve is bent toward a front side of the front panel, and wherein the outdoor unit further comprises: misassembly prevention ribs formed at upper and lower ends of a back surface of the front panel, respectively; and misassembly prevention grooves formed in upper and lower ends of the fastening sleeve, respectively, to accommodate the misassembly prevention ribs, wherein the misassembly prevention ribs and the misassembly prevention grooves are formed at points spaced to a first lateral side or a second lateral side with respect to a vertical line that passes through a center of the through hole.
2. The outdoor unit of claim 1, wherein the orifice comprises a cylindrical body including: a front extension rounded forward from a front end of the body; a close contact portion that extends from a front end of the front extension and configured to come into close contact with a front surface of the front panel; and a rear extension rounded rearward from a rear end of the body, wherein a vortex fence protrudes from an inner circumferential surface of the cylindrical body to be formed in a band shape.
3. The outdoor unit of claim 2, wherein a rear surface of the vortex fence is rounded with a predetermined curvature.
4. The outdoor unit of claim 2, wherein an end of a blade forming the fan is located behind the vortex fence, and wherein an outer diameter of the blade is equal to or greater than an inner diameter of the vortex fence.
5. The outdoor unit of claim 2, wherein coupling guides are formed on the front surface of the front

panel and the close contact portion, respectively, and wherein the coupling guides are disposed on a same vertical line in a state in which the orifice is completely coupled to the front panel.

6. The outdoor unit of claim 1, further comprising: fastening hooks that protrude from a back surface of the close contact portion; and hook grooves formed in the fastening sleeve to accommodate the fastening hooks, wherein when the orifice rotates in a state of being in close contact with the front panel, the fastening hooks are caught in the hook grooves.

7. The outdoor unit of claim 6, wherein the fastening hooks protrude from first and second sides of the back surface of the close contact portion, respectively, wherein the hook grooves are formed in first and second lateral sides of the fastening sleeve, respectively, and wherein the fastening hooks comprise a pair of fastening hooks formed symmetrically with respect to a horizontal plane.